

Physical Chemistry (I)

Lecture 18

Simple Mixtures



Where Are We?

1. Basic concepts
2. The properties of gases
3. The first law
4. Entropy
5. Helmholtz and Gibbs energies
6. Statistical mechanics
7. Phase transitions
8. Simple mixtures
9. Chemical equilibria
10. Electrochemistry

8. Simple Mixtures

Chapter Overview

- Multi-component systems
- Mixing of gases
 - Mixing of perfect gases
 - Mixing of real gases
- Mixing of liquids and solids
 - Ideal solutions
 - Real solutions
 - Liquid-solid solutions: colligative properties

Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_A, n_B, \dots)$
- Thermodynamic potentials
 - $dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$
 - $dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$

Multi-Component System

- Possible phases
 - Each component: gas, liquid, solid
 - Mixed/demixed
- e.g. two-component system (binary system)
 - A: liquid, B: liquid, A-B: mixed
 - A: liquid, B: liquid, A-B: demixed
 - A: gas, B: gas, A-B: mixed
 - ...

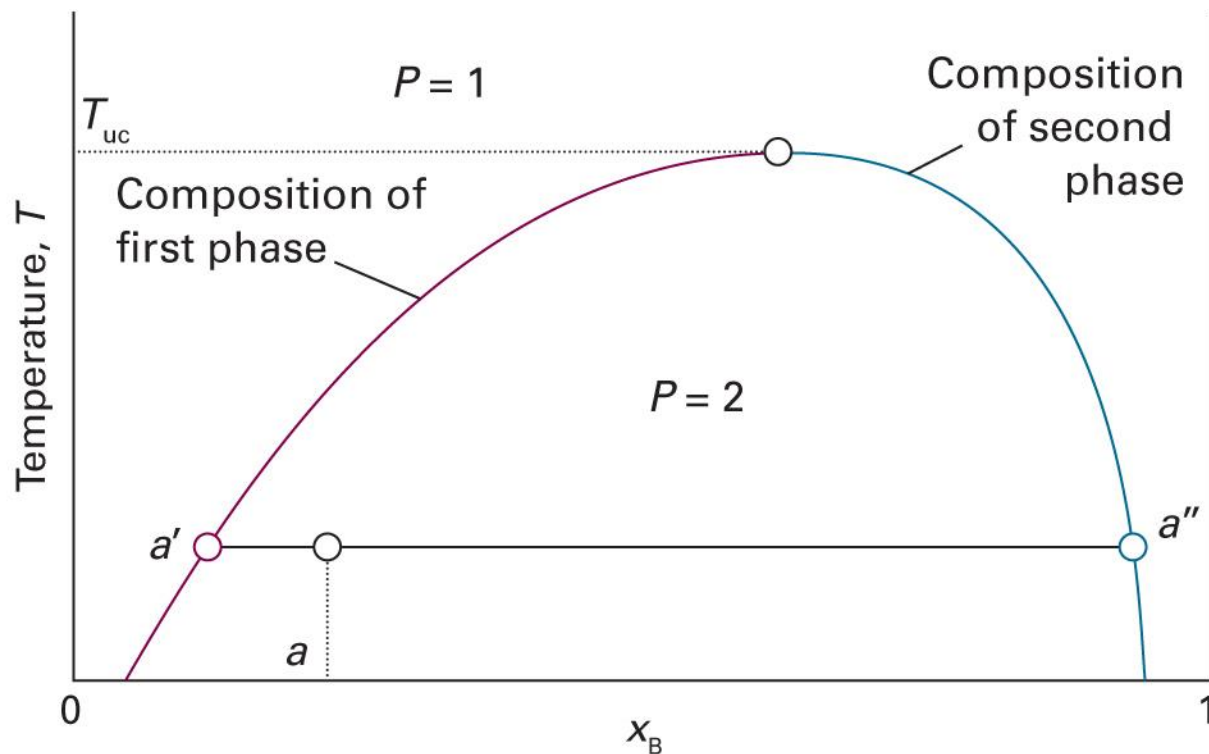
Phase Diagram

- Mole fraction of species J

$$x_J = \frac{n_J}{\sum_J n_J}$$

- At least one axis (usually the x-axis) is for x_J
- The other is for $T, p, \dots$

Example: Binary System



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.13)

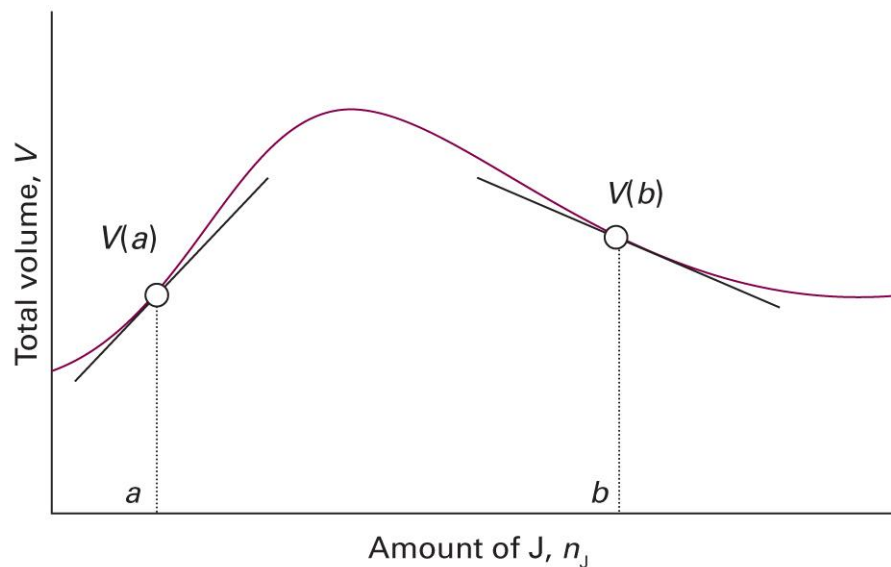
Partial Molar Quantities

$$X_J = \left(\frac{\partial X}{\partial n_J} \right)_{p,T,n'}$$

contribution that a component of a mixture makes
to the total quantity of a sample

Partial Molar Volume

$$V_J = \left(\frac{\partial V}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.2)

Change of Volume

For a two-component system at constant p and T ,

$$V = V(n_A, n_B)$$

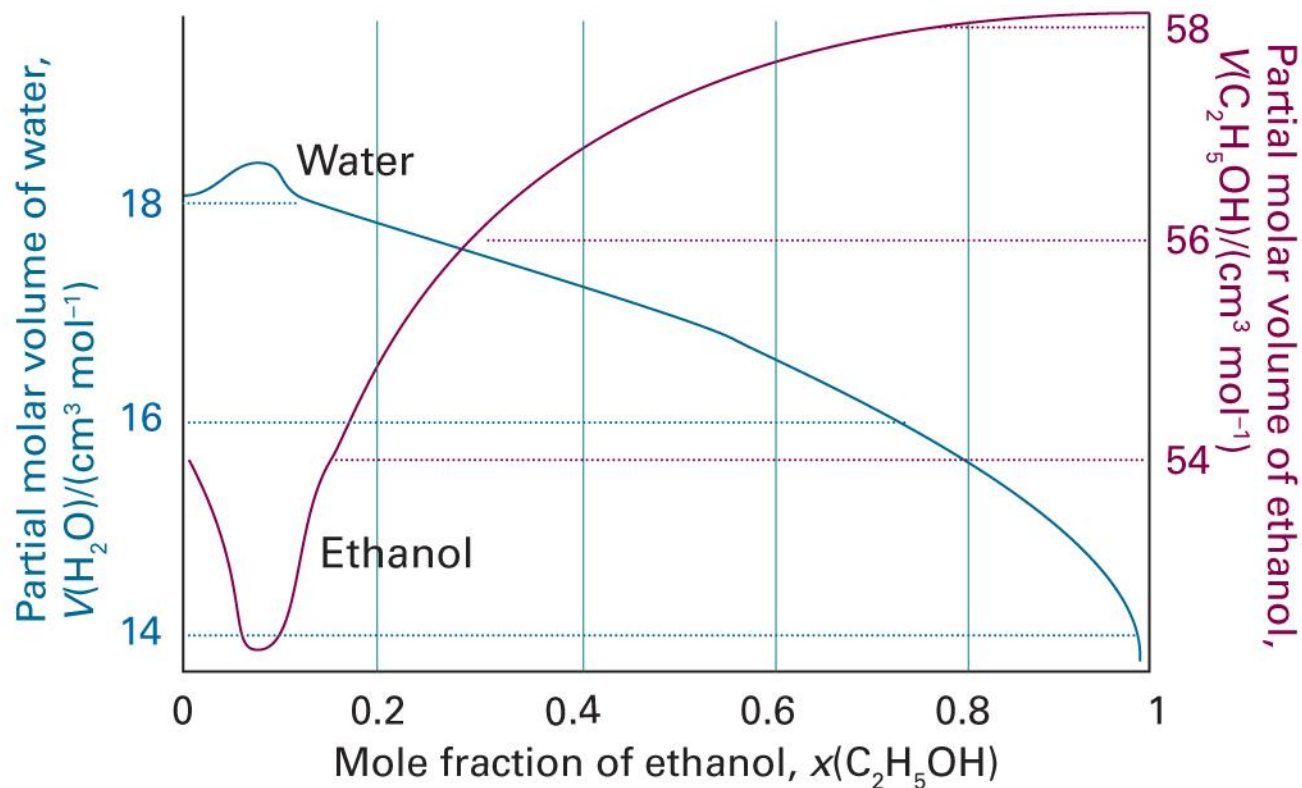
leads to

$$dV = \left(\frac{\partial V}{\partial n_A} \right)_{p,T,n_B} dn_A + \left(\frac{\partial V}{\partial n_B} \right)_{p,T,n_A} dn_B$$

Hence,

$$dV = V_A dn_A + V_B dn_B$$

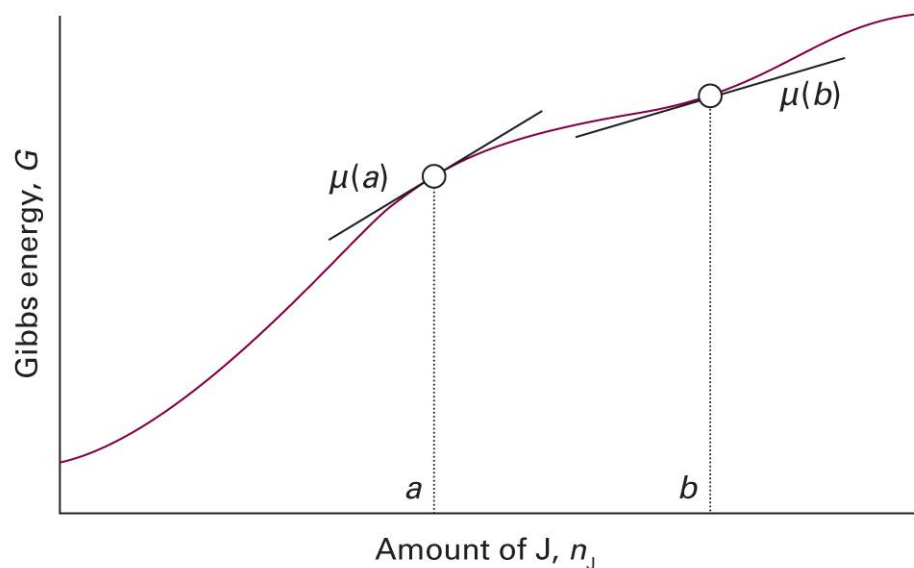
Example: Water + EtOH



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.1)

Partial Gibbs Energy

$$\mu_J = \left(\frac{\partial G}{\partial n_J} \right)_{p,T,n'}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.4)

... is Chemical Potential

$$dU = TdS - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dH = TdS + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dA = -SdT - pdV + \mu_A dn_A + \mu_B dn_B + \dots$$

$$dG = -SdT + Vdp + \mu_A dn_A + \mu_B dn_B + \dots$$

Hence,

$$\mu_J = \left(\frac{\partial U}{\partial n_J} \right)_{S,V,n'} = \left(\frac{\partial H}{\partial n_J} \right)_{S,p,n'} = \left(\frac{\partial A}{\partial n_J} \right)_{T,V,n'} = \left(\frac{\partial G}{\partial n_J} \right)_{T,p,n'}$$

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Be Careful

$$\mu_J = \left(\frac{\partial U}{\partial n_J} \right)_{S,V,n'} = \left(\frac{\partial H}{\partial n_J} \right)_{S,p,n'} = \left(\frac{\partial A}{\partial n_J} \right)_{T,V,n'} = \left(\frac{\partial G}{\partial n_J} \right)_{T,p,n'}$$

$$U_J = \left(\frac{\partial U}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

$$H_J = \left(\frac{\partial H}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

$$A_J = \left(\frac{\partial A}{\partial n_J} \right)_{T,p,n'} \neq \mu_J$$

Gibbs-Duhem Equation

Now, thermodynamic equations for internal energy are

$$U = TS - pV + \sum_J \mu_J n_J$$

$$dU = TdS - pdV + \sum_J \mu_J dn_J$$

Differentiating the first equation,

$$dU = TdS + SdT - pdV - Vdp + \sum_J \mu_J dn_J + \sum_J n_J d\mu_J$$

Comparison leads to

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

At Constant T and p

$$SdT - Vdp + \sum_J n_J d\mu_J = 0$$

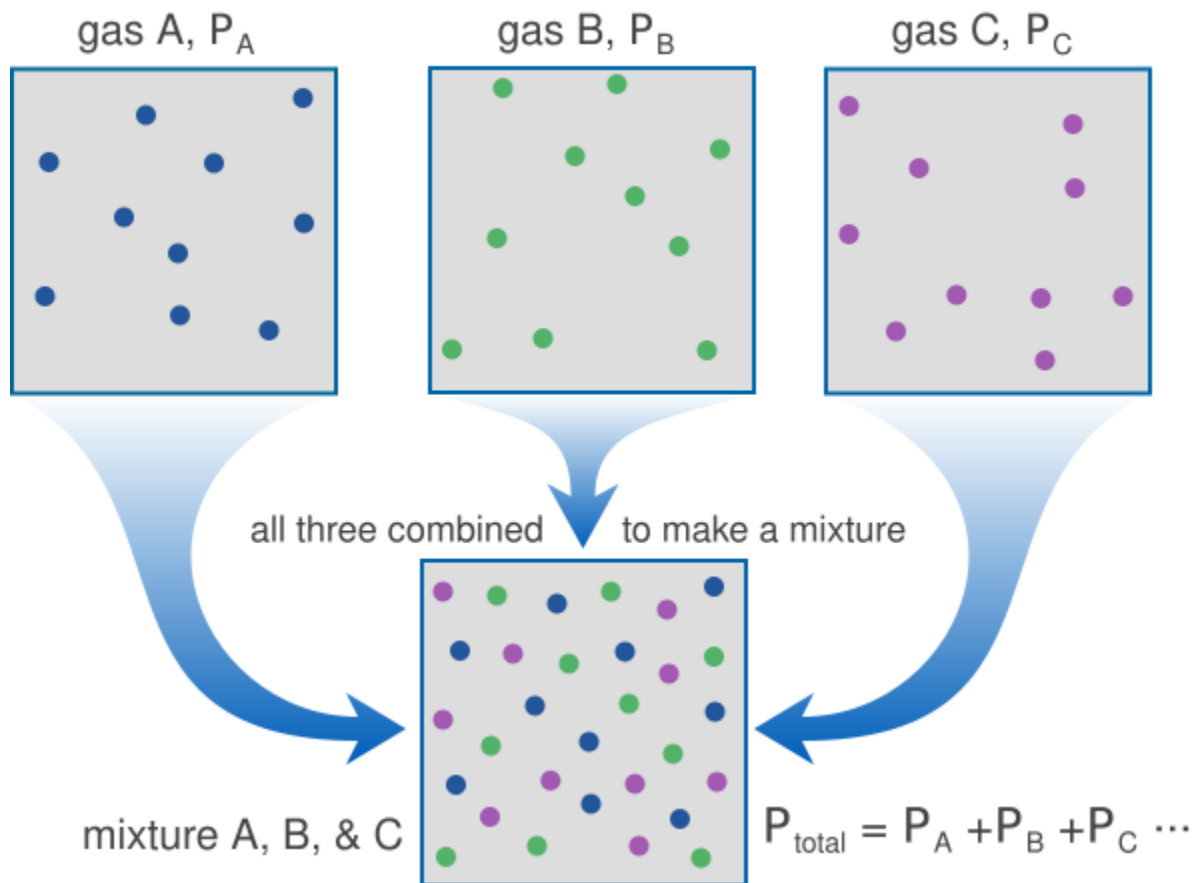
At constant T and p ,

$$\sum_J n_J d\mu_J = 0$$

For a two-component system,

$$n_A d\mu_A + n_B d\mu_B = 0$$

Gas Mixture



Dalton's Law

Partial pressure for species J is

$$p_J = x_J p$$

so

$$p_A + p_B + \cdots = (x_A + x_B + \cdots)p = p$$

Valid for all gases (not only perfect)

Chemical Potential

For a perfect gas,

$$G_m(p) = G_m^\ominus + RT \ln \frac{p}{p^\ominus}$$

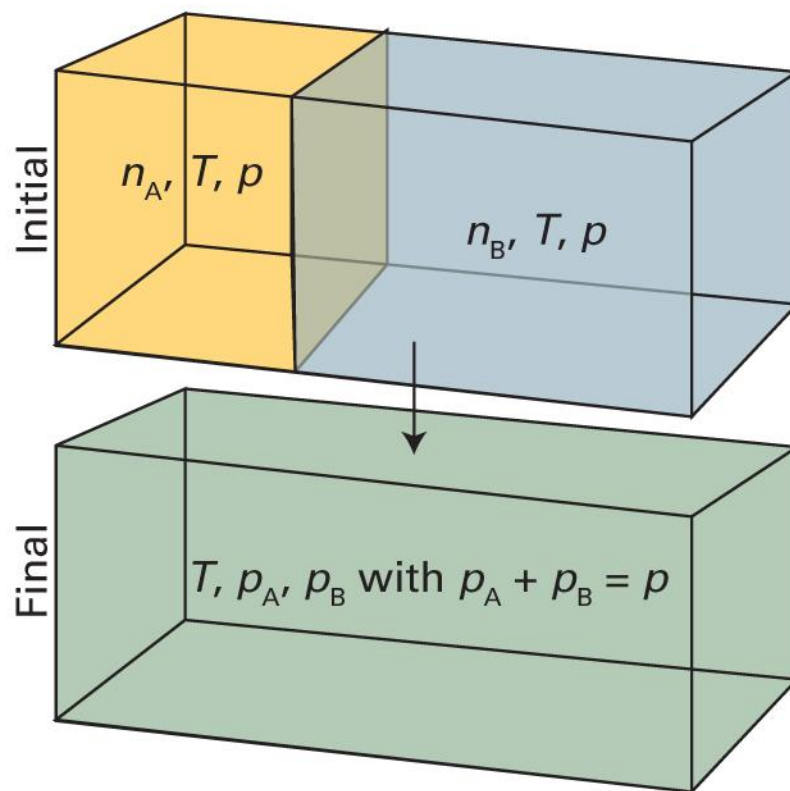
or,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

where $\mu^\ominus$ is the standard chemical potential (at 1 bar).

Mixing of Gases

(at constant T and p)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.6)

Total Gibbs Energy

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

- Initial state

$$G_i = n_A\{\mu_A^\ominus + RT \ln(p/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/p^\ominus)\}$$

- Final state

$$G_f = n_A\{\mu_A^\ominus + RT \ln(p_A/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p_B/p^\ominus)\}$$

Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = G_f - G_i = n_A RT \ln \left(\frac{p_A}{p} \right) + n_B RT \ln \left(\frac{p_B}{p} \right)$$

Using Dalton's law,

$$\Delta_{\text{mix}}G = n_A RT \ln x_A + n_B RT \ln x_B$$

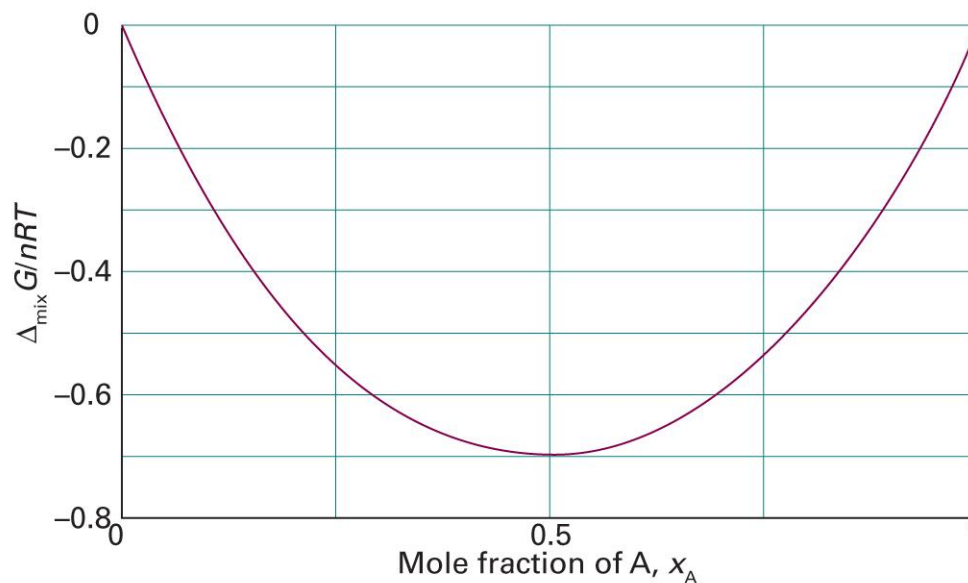
and

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

Note: since $\ln x < 0$ for $x < 1$, $\Delta_{\text{mix}}G < 0$ always.

Gibbs Energy of Mixing

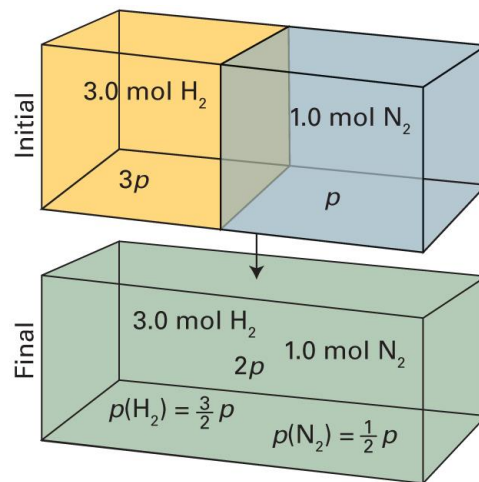
$$\Delta_{\text{mix}}G = nRT\{x_A \ln x_A + (1 - x_A) \ln(1 - x_A)\}$$



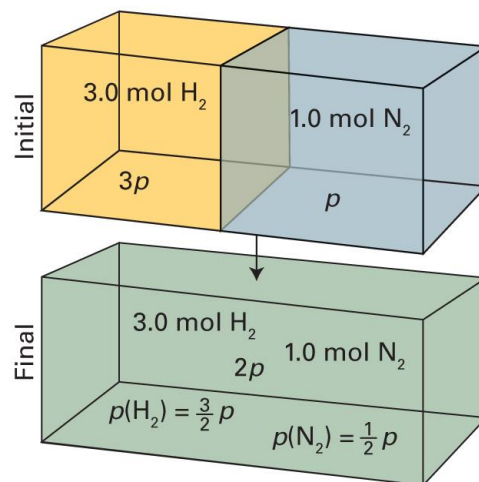
(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.7)

Example 5A.3

A container is divided into two equal compartments. One contains 3.0 mol $\text{H}_2(\text{g})$ at 25°C ; the other contains 1.0 mol $\text{N}_2(\text{g})$ at 25°C . Calculate the Gibbs energy of mixing when the partition is removed. Assume that the gases are perfect.



Example 5A.3



- H₂: amount 3.0 mol, volume $V \rightarrow 2V$, pressure $3p \rightarrow (3/2)p$
- N₂: amount 1.0 mol, volume $V \rightarrow 2V$, pressure $p \rightarrow (1/2)p$

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

Example 5A.3

$$G = G_A + G_B = n_A\mu_A + n_B\mu_B$$

$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$

Total Gibbs energies are

$$G_i = n_A\{\mu_A^\ominus + RT \ln(3p/p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/p^\ominus)\}$$

$$G_f = n_A\{\mu_A^\ominus + RT \ln(3p/2p^\ominus)\} + n_B\{\mu_B^\ominus + RT \ln(p/2p^\ominus)\}$$

Therefore,

$$\Delta_{\text{mix}}G = G_f - G_i = -n_A RT \ln 2 - n_B RT \ln 2 = -(n_A + n_B) RT \ln 2$$

$$= -(3.00 \text{ mol} + 1.00 \text{ mol}) \times (8.3145 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \times 0.693$$

$$= -6.9 \text{ kJ}$$

Entropy and Enthalpy

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

Real Gases

For a perfect gas,

$$\mu = \mu^\ominus + RT \ln \frac{p}{p^\ominus}$$

For a real gas,

$$\mu = \mu^\ominus + RT \ln \frac{f}{p^\ominus}$$

where f is the **fugacity**.

Properties of Fugacity

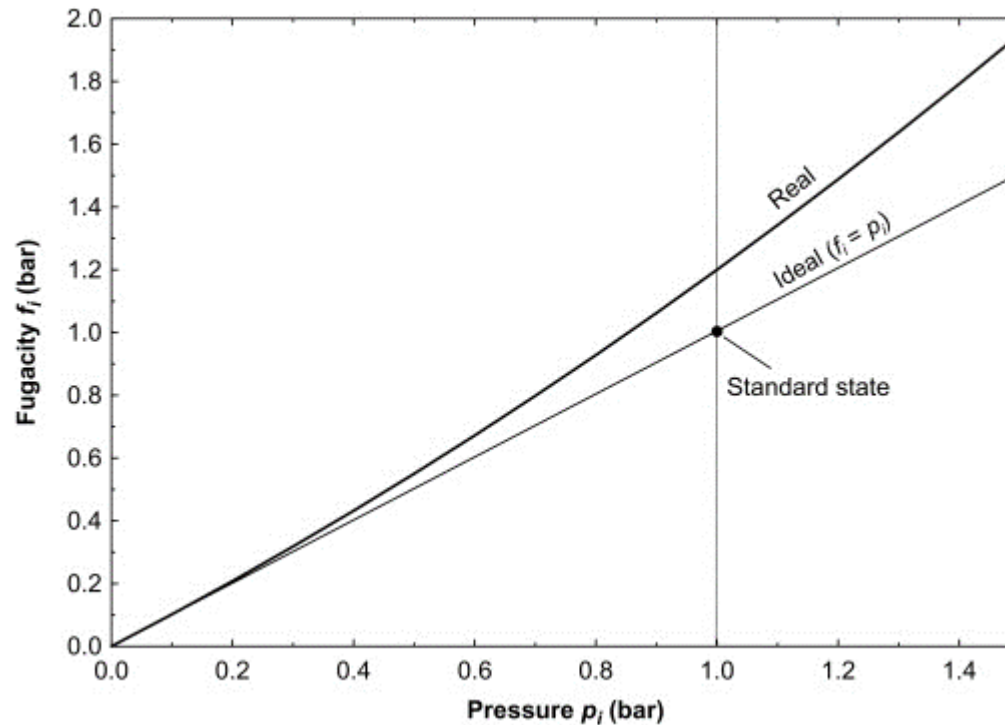
- As $p \rightarrow 0$, $f \rightarrow p$
- Fugacity coefficient $\phi = f/p$
- Molecular interpretation
 - $f > p$ or $\phi > 1$: repulsion-dominant
 - $f = p$ or $\phi = 1$: perfect-gas-like
 - $f < p$ or $\phi < 1$: attraction-dominant

NOTE

$$\begin{aligned}\mu &= \mu^\ominus + \int_{p^\ominus}^p V_m \, dp \\ &= \mu^\ominus + \int_{p^\ominus}^p \frac{RT}{p} \, dp \\ &= \mu^\ominus + RT \ln \frac{p}{p^\ominus}\end{aligned}$$

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Fugacity and Pressure



Mixing of Liquids

$$\mu_A(l) = \mu_A(g)$$

When A is a pure substance,

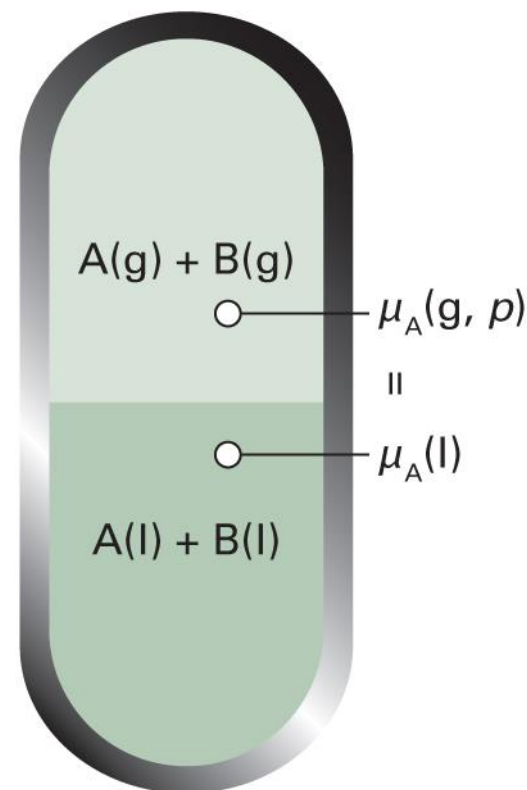
$$\underbrace{\mu_A^*(l)}_{\text{liquid}} = \underbrace{\mu_A^\ominus(g) + RT \ln(p_A^*/p^\ominus)}_{\text{(perfect) gas}}$$

If A is mixed with another substance B,

$$\mu_A(l) = \mu_A^\ominus(g) + RT \ln(p_A/p^\ominus)$$

Subtraction leads to

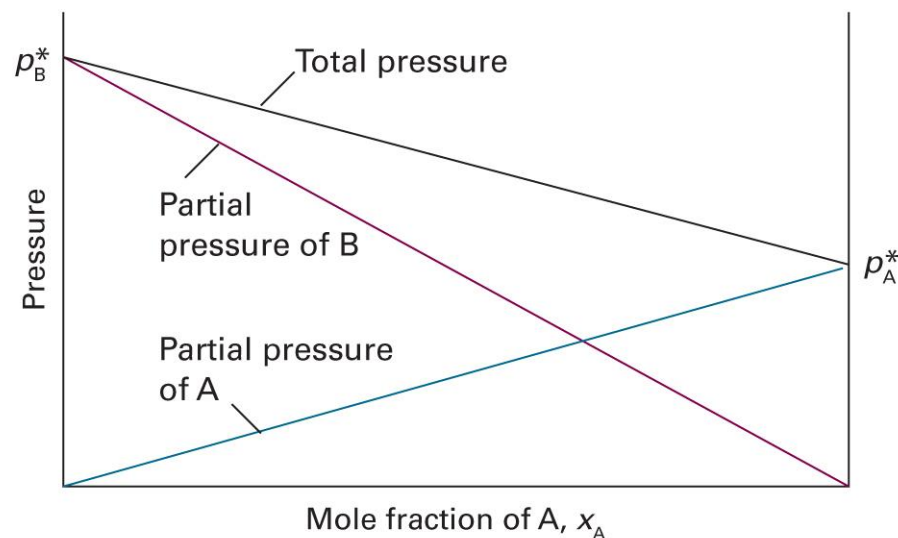
$$\mu_A(l) = \mu_A^*(l) + RT \ln(p_A/p_A^*)$$



Raoult's Law: Ideal Solution

$$p_A = x_A p_A^*$$

- **Ideal solutions:** mixtures that obey the law throughout the composition range from pure A to pure B



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5A.11)

Chemical Potential

$$\mu_A(l) = \mu_A^*(l) + RT \ln(p_A/p_A^*)$$

For an ideal solution,

$$p_A = x_A p_A^*$$

Hence,

$$\mu_A(l) = \mu_A^*(l) + RT \ln x_A$$

As $x_A < 1$, $\mu_A(l) < \mu_A^*(l)$ and mixed phase is more stable

Ideal Solutions

- Before mixing:

$$G_i = n_A \mu_A^* + n_B \mu_B^*$$

- After mixing:

$$G_f = n_A \mu_A + n_B \mu_B$$

$$= n_A (\mu_A^* + RT \ln x_A) + n_B (\mu_B^* + RT \ln x_B)$$

- Gibbs energy of mixing

$$\Delta_{\text{mix}} G = G_f - G_i = n_A RT \ln x_A + n_B RT \ln x_B$$

$$\Delta_{\text{mix}} G = nRT(x_A \ln x_A + x_B \ln x_B)$$

Entropy and Enthalpy

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B)$$

- Entropy of mixing

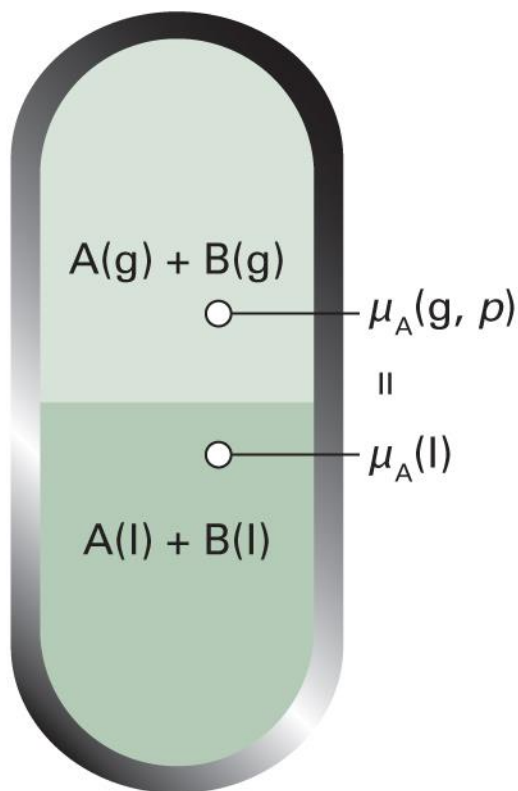
$$\Delta_{\text{mix}}S = -\left(\frac{\partial \Delta_{\text{mix}}G}{\partial T}\right)_p = -nR(x_A \ln x_A + x_B \ln x_B)$$

- Enthalpy of mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T\Delta_{\text{mix}}S = 0$$

(Same with perfect gas mixtures)

Liquid-Vapor Equilibrium



- Constraints

$$\mu_A(g) = \mu_A(l)$$

$$\mu_B(g) = \mu_B(l)$$

- May have $x_A(g) \neq x_A(l)$

Brief Illustration 5A.3

Consider a mixture of benzene and methylbenzene, which form an approximately ideal solution. The vapour pressure of pure benzene at 20°C is 75.0 Torr and that of pure methylbenzene is 21.0 Torr at the same temperature. What are the mole fractions in the vapour when the two liquids are in an equimolar mixture?

In an equimolar mixture,

$$x_{\text{benzene}}(l) = x_{\text{methylbenzene}}(l) = 0.5$$

Thus,

$$p_{\text{benzene}} = 75.0 \text{ Torr} \times 0.5 = 37.5 \text{ Torr}$$

$$p_{\text{methylbenzene}} = 21.0 \text{ Torr} \times 0.5 = 10.5 \text{ Torr}$$

Brief Illustration 5A.3

The total pressure is $37.5 \text{ Torr} + 10.5 \text{ Torr} = 48.0 \text{ Torr}$.

From Dalton's law,

$$x_{\text{benzene}}(g) = (37.5 \text{ Torr}) / (48.0 \text{ Torr}) = 0.78$$

$$x_{\text{methylbenzene}}(g) = (10.5 \text{ Torr}) / (48.0 \text{ Torr}) = 0.22$$

Vapor Pressures

From Raoult's law,

$$p_A = x_A p_A^*, \quad p_B = x_B p_B^*$$

The total vapor pressure is

$$p = p_A + p_B = x_A p_A^* + x_B p_B^* = x_A p_A^* + (1 - x_A) p_B^*$$

which becomes

$$p = p_B^* + (p_A^* - p_B^*) x_A$$

Vapor Compositions

$$p = p_B^* + (p_A^* - p_B^*)x_A$$

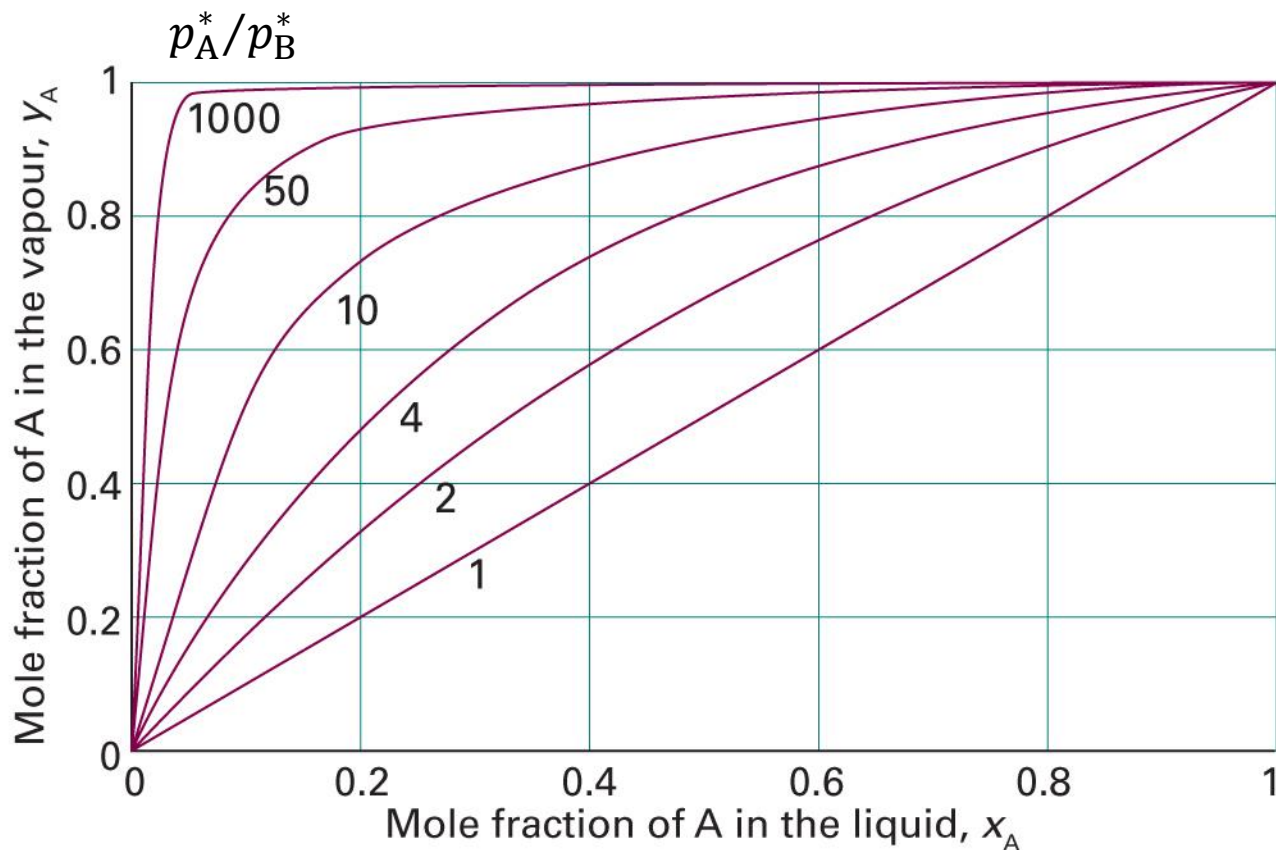
Denote the mole fractions in the vapor as y_A and y_B :

$$y_A = p_A/p, \quad y_B = p_B/p$$

Now,

$$y_A = \frac{x_A p_A^*}{p_B^* + (p_A^* - p_B^*)x_A}, \quad y_B = 1 - y_A$$

Vapor Compositions



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5C.2)

x_A as a Function of y_A

$$y_A = \frac{x_A p_A^*}{p_B^* + (p_A^* - p_B^*)x_A}$$

$$p_B^* y_A + (p_A^* - p_B^*) x_A y_A = x_A p_A^*$$

$$p_B^* y_A = x_A \{p_A^* - (p_A^* - p_B^*) y_A\}$$

Hence,

$$x_A = \frac{y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

Total Vapor Pressure

$$x_A = \frac{y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

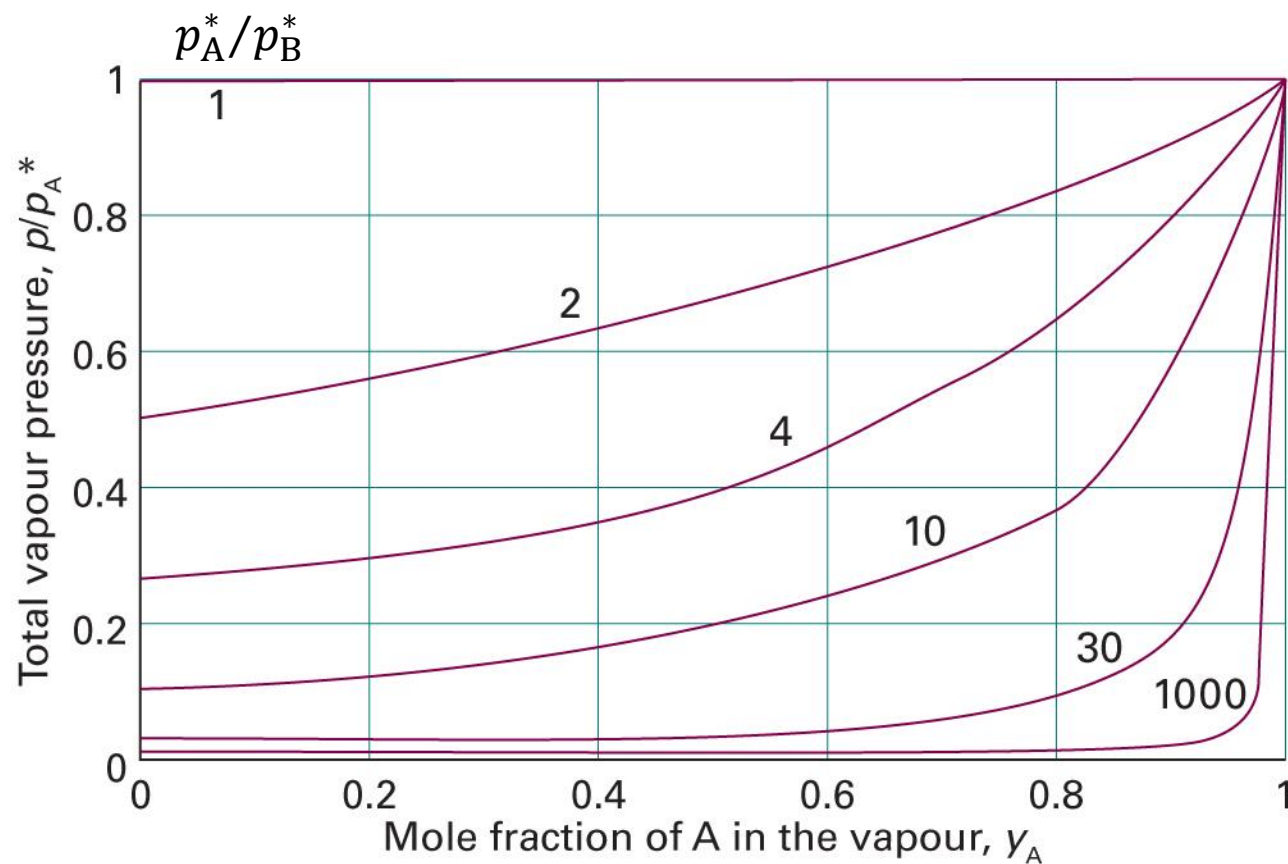
The total vapor pressure is

$$p = p_B^* + (p_A^* - p_B^*) x_A = p_B^* + \frac{(p_A^* - p_B^*) y_A p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

which leads to

$$p = \frac{p_A^* p_B^*}{p_A^* + (p_B^* - p_A^*) y_A}$$

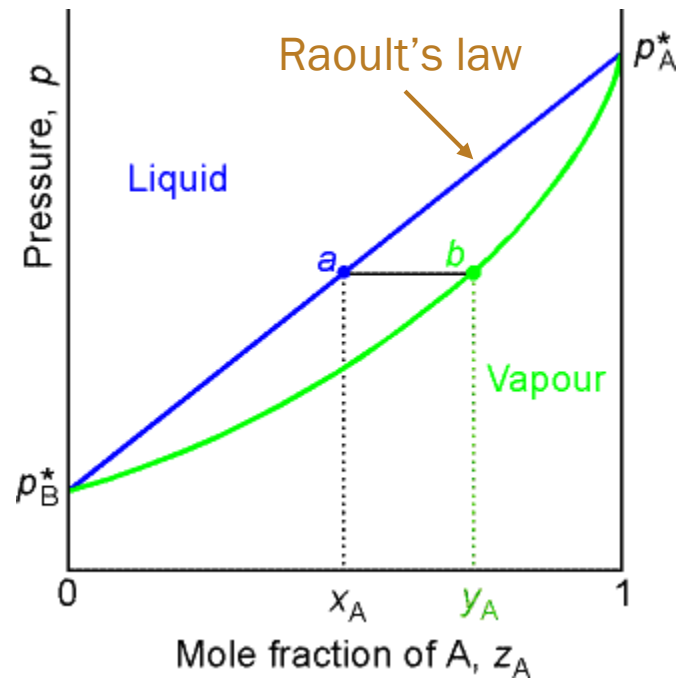
Total Vapor Pressure



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 5C.3)

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Pressure-Composition Diagram



(Image: <http://people.cst.cmich.edu/teckl1mm/pchemi/chm351ch8af01.htm>)